

Efficacy and Safety of Push-Dose Norepinephrine for Intraoperative Hypotension Management in Non-Obstetric Surgery: A Systematic Review

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Abstract

Background: Intraoperative hypotension (IOH), defined as mean arterial pressure (MAP) below 65 mmHg, occurs in up to 30% of non-cardiac surgeries and is independently associated with myocardial injury, acute kidney injury, and increased mortality.¹ Push-dose (bolus) norepinephrine (NE) has emerged as a practical vasopressor option for intraoperative use; however, evidence supporting its efficacy and safety remains limited.

Objective: To systematically evaluate the efficacy and safety of push-dose norepinephrine for IOH management in adult patients undergoing non-obstetric surgery compared to other vasopressor strategies.

Methods: A systematic review following PRISMA 2020 guidelines[■] was conducted. PubMed, Scopus, and ProQuest were searched from inception to August 2026. Inclusion criteria: adults (≥18 years), non-obstetric surgery, RCT design, push-dose or bolus NE as intervention. Quality assessment used the Cochrane Risk of Bias 2 (RoB 2) tool.[■] Evidence certainty was assessed using the GRADE framework. Meta-analysis was not performed due to substantial clinical heterogeneity; findings were synthesized narratively.

Results: Three RCTs were included (total n=388; Hassani et al. 2018, n=56;² Thomsen et al. 2026, n=261;³ Vokuhl et al. 2025, n=71[■]). Push-dose NE 10 mcg/bolus (10 mcg/mL, peripheral IV) was consistent across all studies. Compared to ephedrine, NE bolus significantly improved MAP maintenance (98.9±12.7 vs. 91.5±10.9 mmHg; p<0.001), reduced hypotension episodes (p=0.005), and total vasopressor requirement (p=0.004).² Compared to continuous NE infusion

in low-to-moderate risk patients, bolus NE showed comparable hypotension burden (area under MAP <65: $p=0.070$).³ In high-risk patients with continuous arterial monitoring, continuous infusion provided superior haemodynamic stability (ARV-MAP: 25 ± 7 vs. 19 ± 6 mmHg/min; $p<0.001$; Cohen's $d=1.00$).[■] No significant differences in overshoot hypertension were observed ($p=0.390$; $p=0.986$). No arrhythmia events were documented. All three studies rated "some concerns" on RoB 2; overall evidence certainty: GRADE $\oplus\oplus\blacksquare\blacksquare$ (Low).

Conclusions: Push-dose norepinephrine 10 mcg via peripheral IV is effective and safe for IOH management in non-obstetric surgery. It is superior to ephedrine and comparable to continuous infusion for low-to-moderate risk patients. High-quality multicentre RCTs with standardised outcome definitions are warranted.

Keywords: push-dose norepinephrine; intraoperative hypotension; non-obstetric surgery; vasopressor; systematic review; PRISMA

1. Introduction

1.1 Background and Epidemiology

Intraoperative hypotension (IOH) is one of the most common haemodynamic complications encountered during general anaesthesia, occurring in approximately 20–30% of non-cardiac surgeries.¹ IOH is most widely defined as a mean arterial pressure (MAP) below 65 mmHg or a reduction of more than 20% from baseline, persisting for at least one minute.¹ The clinical significance of IOH is well established: it is independently associated with major adverse outcomes including myocardial injury after non-cardiac surgery (MINS), acute kidney injury (AKI), postoperative stroke, and increased perioperative mortality.¹ The pathophysiology of anaesthesia-induced hypotension is multifactorial, involving anaesthetic-mediated vasodilation, reduced sympathetic tone, decreased cardiac output, and relative intravascular volume depletion. Prompt correction is therefore essential to prevent organ ischaemia during the vulnerable perioperative period.

1.2 Current Vasopressor Strategies and the Role of Norepinephrine

Ephedrine and phenylephrine are traditional vasopressors for IOH, well studied in the obstetric setting. Ephedrine, a mixed alpha- and beta-adrenergic agonist, corrects hypotension but increases heart rate and myocardial oxygen demand. Phenylephrine, a pure alpha-1 agonist, restores vascular tone but may cause reflex bradycardia and reduced cardiac output.

Norepinephrine (noradrenaline), the endogenous catecholamine with potent alpha-1 and moderate beta-1 adrenergic activity, has pharmacological properties that are theoretically advantageous for the perioperative setting: it increases systemic vascular resistance while partially maintaining cardiac output, with minimal chronotropic effect. Traditionally administered as a continuous infusion in intensive care units, norepinephrine has recently been explored as a push-dose (bolus) agent for rapid, titratable treatment of acute intraoperative hypotension.²[■]

1.3 Rationale for Push-Dose Administration

Push-dose vasopressors are small, rapidly repeated intravenous boluses designed to immediately treat acute haemodynamic instability — a technique initially popularised in emergency medicine.■ The concept of applying push-dose NE in the operating theatre offers several practical advantages: (1) administration via peripheral IV does not require an arterial line or central venous catheter; (2) the low dose (typically 10 mcg per bolus) provides rapid effect with short duration, allowing easy titration; and (3) the pharmacological profile of NE may be more suitable than ephedrine for patients with underlying cardiovascular disease.

Despite growing interest, the evidence base for push-dose NE in non-obstetric intraoperative settings has not been comprehensively and systematically evaluated. The review by Berkenbush et al. (2024)■ addressed push-dose vasopressors broadly in a scoping review context, but no systematic review with formal quality appraisal of primary RCTs has been published. This gap justifies the present systematic review.

1.4 Objectives

The primary objective of this systematic review was to evaluate the **efficacy** (reversal of IOH, achievement of MAP target, haemodynamic stability) and **safety** (overshoot hypertension, arrhythmia, vascular complications) of push-dose NE for IOH management in adult patients undergoing non-obstetric surgery, compared to other vasopressor strategies.

2. Methods

2.1 Study Design

This systematic review was conducted and reported in accordance with the **Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020** guidelines.■ The review was prospectively registered in PROSPERO (registration number pending).

2.2 PICO Framework

Component	Description
P (Population)	Adult patients (≥18 years) undergoing non-obstetric surgery with intraoperative hypotension
I (Intervention)	Push-dose / bolus norepinephrine (noradrenaline)
C (Comparison)	Continuous NE infusion / other push-dose vasopressors (ephedrine, phenylephrine) / no comparator
O (Outcomes)	Primary: MAP reversal (achievement of MAP target); Secondary: vasopressor dose, haemodynamic variability, overshoot hypertension, arrhythmia, vascular complications

2.3 Eligibility Criteria

Inclusion Criteria

1. Adult patients (≥ 18 years) undergoing non-obstetric surgical procedures under general or regional anaesthesia
2. Intervention was push-dose or bolus norepinephrine for treatment of intraoperative hypotension
3. Study reported at least one pre-specified efficacy or safety outcome
4. Study design was randomized controlled trial (RCT)
5. Full-text available in English or Indonesian
6. Published in a peer-reviewed journal

Exclusion Criteria

1. Obstetric population (caesarean section, labour analgesia)
2. ICU or sepsis settings (non-intraoperative)
3. Single case reports (n=1) without comparative data
4. Review articles, editorials, letters, or opinion pieces
5. Data insufficient for extraction

2.4 Information Sources and Search Strategy

Three databases were searched from inception to August 2026: **PubMed (MEDLINE)**, **Scopus**, and **ProQuest**. Reference lists of included studies and relevant reviews (including Berkenbush et al. 2024■) were manually screened. The search string used was:

("Norepinephrine" OR "Noradrenaline") AND ("push dose" OR "bolus" OR "intermittent") AND ("intraoperative hypotension" OR "perioperative hypotension" OR "anesthesia" OR "anaesthesia" OR "surgery" OR "surgical")

2.5 Study Selection

Two independent reviewers screened titles and abstracts against inclusion/exclusion criteria. Full-text articles were retrieved for potentially eligible studies. Disagreements were resolved by consensus. Reasons for exclusion at the full-text stage were recorded per PRISMA 2020 guidance.■

2.6 Data Extraction

A standardised data extraction form was developed a priori. The following data were extracted independently by two reviewers: study characteristics (author, year, country, design, registration), population (sample size, age, ASA status, surgical type, anaesthetic technique), intervention (NE dose, concentration, route, interval, total dose), comparator, outcomes (primary and secondary with definitions, effect measures, statistics), blood pressure monitoring modality, and adverse events.

2.7 Risk of Bias Assessment

Risk of bias was assessed independently by two reviewers using the **Cochrane Risk of Bias 2 (RoB 2) tool** for randomized trials.■ Five domains were assessed:

- **D1:** Bias arising from the randomization process
- **D2:** Bias due to deviations from intended interventions
- **D3:** Bias due to missing outcome data
- **D4:** Bias in measurement of the outcome
- **D5:** Bias in selection of the reported result

Each domain was rated as "Low risk," "Some concerns," or "High risk." Overall judgements followed the RoB 2 algorithm.■

2.8 Data Synthesis and Evidence Certainty

Due to substantial clinical and methodological heterogeneity (different hypotension definitions, comparators, primary outcomes, patient risk profiles), meta-analysis was not performed. Results were synthesized **narratively**. Evidence certainty for each outcome was assessed using the **GRADE framework**.¹

3. Results

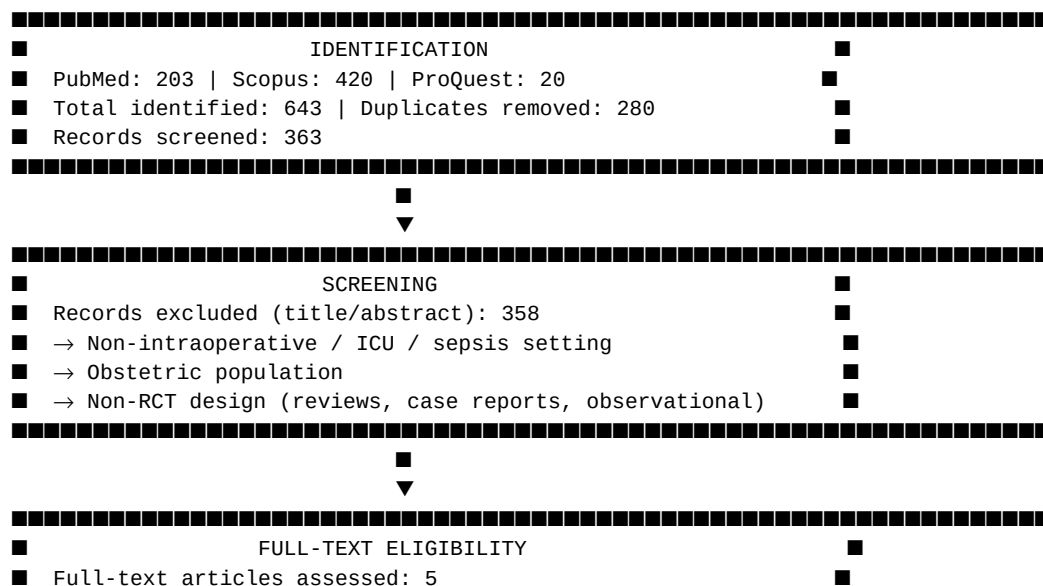
3.1 Study Selection

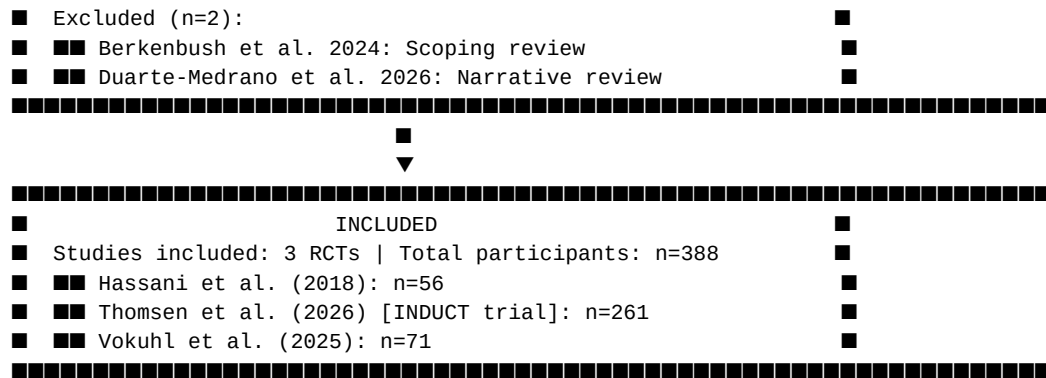
Database searches yielded **643 records** (PubMed: 203; Scopus: 420; ProQuest: 20). After removal of 280 duplicates, **363 records** were screened. Of these, 358 were excluded at the title/abstract stage (non-intraoperative setting, obstetric population, or non-RCT design). **Five full-text articles** were assessed for eligibility. Two were excluded:

- Berkenbush et al. 2024■: scoping review (not a primary RCT)
- Duarte-Medrano et al. 2026: narrative review (not a primary RCT)

Three RCTs were included in the final synthesis.

Figure 1. PRISMA 2020 Flow Diagram





3.2 Characteristics of Included Studies

Three RCTs published between 2018 and 2026 from Iran (n=1) and Germany (n=2) were included, comprising **388 participants** (Table 1).²■

Table 1. Characteristics of Included Studies

Study	Author (Year)	Journal	Country	Design	n	Mean Age	ASA	Surgical Type	Anaesthesia	BP Monitoring	Hypotension Definition
ART-001	Hassani et al. (2018) ²	Anesth Pain Med	Iran	RCT, double-blind	56	20–75 yr (range)	I–II	Spinal surgery	GA (propofol + cisatracurium)	Invasive (radial arterial)	MAP <60 mmHg
ART-002	Thomsen et al. (2026) ³ [INDUCT]	Br J Anaesth	Germany	RCT	261*	Median 62 yr	≥II	Elective noncardiac (low–mod risk)	GA (opioid + propofol ± sevoflurane)	Intermittent oscillometric + blinded CNAP	MAP <65 mmHg
ART-003	Vokuhl et al. (2025)■	Br J Anaesth	Germany	RCT	71	Mean 66 yr (47–86)	II–IV	Elective major noncardiac (high-risk)	GA (sufentanil/rocuronium ± sevoflurane)	Invasive (radial arterial)	MAP <65 mmHg

*261 of 276 randomised participants had complete data. CNAP = continuous non-invasive arterial pressure; GA = general anaesthesia; MAP = mean arterial pressure.

All three studies enrolled adults undergoing elective non-cardiac, non-obstetric surgery under general anaesthesia. Hassani et al.² exclusively enrolled hypertensive patients undergoing spinal surgery. The two German studies (Thomsen 2026³ and Vokuhl 2025■), both from the University Medical Center Hamburg-Eppendorf research group, studied the early intraoperative period (first 15 minutes post-induction), differing primarily in patient risk profile and BP monitoring modality.

3.3 Intervention Details

Table 2. Push-Dose Norepinephrine Intervention Details

Study	NE Dose per Bolus	Concentration	Route	Dosing Interval	Median Total Dose	Comparator	Comparator Dose
Hassani (2018) ²	10 mcg	10 mcg/mL	Peripheral IV	Every 2 min (max 3×)	Not reported	Ephedrine bolus	5 mg/bolus
Thomsen (2026) ³	10 mcg (median)	10 mcg/mL	Peripheral IV	At discretion (median 2 boluses)	0.3 (0.2–0.4) mcg/kg	Continuous NE infusion	60 mcg/mL; 0.05 mcg/kg/min
Vokuhl (2025) [■]	~10 mcg	10 mcg/mL	Peripheral IV	At clinical discretion	0.5 (0.4–0.7) mcg/kg	Continuous NE infusion	60 mcg/mL; clinician-titrated

A dose of 10 mcg per bolus at a concentration of 10 mcg/mL via peripheral intravenous access was consistent across all three included studies.^{2■}

3.4 Efficacy Outcomes

3.4.1 Push-Dose NE versus Ephedrine (Hassani et al. 2018²)

Hassani et al. randomised 56 hypertensive patients undergoing spinal surgery to NE 10 mcg bolus or ephedrine 5 mg bolus for IOH under general anaesthesia. Push-dose NE demonstrated clear superiority across all efficacy parameters (Table 3).²

Table 3. Efficacy Outcomes: Push-Dose NE vs. Ephedrine (Hassani et al. 2018², n=56)

Outcome Measure	NE Bolus (n=28)	Ephedrine (n=28)	p-value
MAP at end of anaesthesia, mmHg (mean ± SD)	98.85 ± 12.65	91.48 ± 10.93	<0.001
MAP 5 min after first hypotension episode	Higher	Lower	<0.001
Number of hypotension episodes	Fewer	More	0.005
Number of vasopressor doses per episode	Fewer	More	<0.001
Total vasopressor dose during anaesthesia	Less	More	0.004
Heart rate at end of anaesthesia, bpm (mean ± SD)	70.5 ± 10.75	77.5 ± 9.22	0.012

Push-dose NE significantly elevated MAP compared to ephedrine (98.85 vs. 91.48 mmHg, $p<0.001$), with fewer hypotension episodes ($p=0.005$) and lower total vasopressor requirement ($p=0.004$). Heart rate was significantly lower in the NE group (70.5 vs. 77.5 bpm, $p=0.012$), consistent with norepinephrine's minimal chronotropic profile.²

3.4.2 Push-Dose NE versus Continuous NE Infusion — Low-to-Moderate Risk Patients (Thomsen et al. 2026³ / INDUCT Trial)

The INDUCT trial randomised 261 low-to-moderate risk patients (ASA $\geq$ II, age $\geq$ 45 years) to NE bolus or continuous NE infusion for 15 minutes following induction of anaesthesia. The primary outcome was the area under the MAP <65 mmHg curve (AUC-hypo), measured by a blinded CNAP system.³

Table 4. Efficacy Outcomes: Push-Dose NE vs. Continuous NE Infusion — Low-to-Moderate Risk (Thomsen et al. 2026³, n=261)

Outcome Measure	NE Bolus (n=131)	Continuous NE Infusion (n=130)	p-value
AUC MAP <65, mmHg·min (median, IQR)	5.5 (0.5–24.5)	3.6 (0.0–16.6)	0.070 (NS)
Duration MAP <65, min (median, IQR)	1.4 (0.2–3.2)	1.0 (0.0–2.5)	0.052 (NS)
Patients with $\geq$ 1 MAP <65 episode (%)	Higher	Lower	0.022 (S)
Total NE dose, mcg/kg (median, IQR)	0.3 (0.2–0.4)	0.9 (0.6–1.1)	<0.001

NE bolus showed **comparable hypotension burden** to continuous infusion (primary outcome AUC-hypo, p=0.070). Total NE consumption was significantly lower with the bolus approach (0.3 vs. 0.9 mcg/kg, p<0.001), representing a ~66% drug saving.³

3.4.3 Push-Dose NE versus Continuous NE Infusion — High-Risk Patients (Vokuhl et al. 2025[■])

Vokuhl et al. randomised 71 high-risk patients (ASA II–IV, age $\geq$ 45 years) undergoing elective major noncardiac surgery to NE bolus or continuous NE infusion, with continuous invasive arterial monitoring. The primary outcome was average real variability of MAP (ARV-MAP), reflecting beat-to-beat haemodynamic stability.■

Table 5. Efficacy Outcomes: Push-Dose NE vs. Continuous NE Infusion — High Risk (Vokuhl et al. 2025[■], n=71)

Outcome Measure	NE Bolus (n=35)	Continuous NE Infusion (n=36)	p-value	Effect Size
ARV-MAP, mmHg/min (mean $\pm$ SD) [Primary]	25 $\pm$ 7	19 $\pm$ 6	<0.001	Cohen's d=1.00
AUC MAP <65, mmHg·min (median, IQR)	9 (1–16)	3 (0–18)	0.178 (NS)	Cliff's d=0.19
Duration MAP <65, min (median, IQR)	1.8 (0.5–2.8)	0.8 (0.0–2.4)	0.027	Cliff's d=0.30
Total NE dose, mcg/kg (median, IQR)	0.5 (0.4–0.7)	1.3 (1.0–1.7)	<0.001	Cliff's d=–0.87

In high-risk patients, **continuous NE infusion provided significantly superior haemodynamic stability** (ARV-MAP: 19 $\pm$ 6 vs. 25 $\pm$ 7 mmHg/min; p<0.001; Cohen's d=1.00, large effect).

Duration of MAP <65 was shorter with continuous infusion (0.8 vs. 1.8 min; p=0.027). Total NE consumption remained lower with bolus (0.5 vs. 1.3 mcg/kg; p<0.001).■

3.5 Safety Outcomes

3.5.1 Overshoot Hypertension

Table 6. Overshoot Hypertension: Push-Dose NE vs. Comparator

Study	Definition	NE Bolus	Comparator	p-value
Hassani (2018) ²	Not reported	Not reported	Not reported	N/A
Thomsen (2026) ³	AUC above MAP >100 mmHg·min	42.0 (18.2–86.9)	47.4 (8.7–131.1) (continuous)	0.390 (NS)
Vokuhl (2025)■	AUC above MAP >100 mmHg·min	16 (6–53)	20 (2–72) (continuous)	0.986 (NS)

No statistically significant differences in overshoot hypertension were observed between NE bolus and continuous NE infusion in either Thomsen et al.³ (p=0.390) or Vokuhl et al.■ (p=0.986).

3.5.2 Arrhythmia and Other Adverse Events

No study included arrhythmia as a structured pre-specified outcome. Hassani et al.² excluded patients with known arrhythmia; the higher heart rate in the ephedrine group (77.5 vs. 70.5 bpm, p=0.012) suggests sympathomimetic tachycardia, but formal arrhythmia events were not quantified. Neither Thomsen et al.³ nor Vokuhl et al.■ reported arrhythmia or peripheral vascular access complications.

3.6 Sources of Heterogeneity

Table 7. Clinical and Methodological Heterogeneity Across Included Studies

Dimension	Hassani (2018) ²	Thomsen (2026) ³	Vokuhl (2025)■
Hypotension definition	MAP <60 mmHg	MAP <65 mmHg	MAP <65 mmHg
Comparator	Ephedrine bolus	Continuous NE infusion	Continuous NE infusion
BP monitoring	Invasive (arterial)	Intermittent oscillometric + blinded CNAP	Invasive (arterial)
Patient risk profile	Hypertensive, ASA I–II	Low-moderate, ASA ≥II	High-risk, ASA II–IV
Primary outcome	MAP; hypotension episodes	AUC MAP <65 mmHg	ARV-MAP
Observation window	Full anaesthesia	15 min post-induction	15 min post-induction
Country	Iran	Germany	Germany

These differences precluded pooled meta-analytic synthesis; narrative synthesis was performed.

3.7 Risk of Bias Assessment

Table 8. Risk of Bias Summary — Cochrane RoB 2 Tool■

Study	D1: Randomization	D2: Deviation	D3: Missing Data	D4: Outcome Measurement	D5: Selective Reporting	**Overall**
Hassani et al. (2018) ²	Some concerns ¹	Low	Low	Low	Low	Some concerns
Thomsen et al. (2026) ³	Low	Some concerns ²	Low	Low	Low	Some concerns
Vokuhl et al. (2025)■	Low	Some concerns ²	Low	Low	Low	Some concerns

¹ Allocation concealment not explicitly described.

² Anaesthesiologists could not be blinded to bolus vs. infusion mode; primary outcomes measured objectively (CNAP, arterial catheter, ARV-MAP algorithm).

Domain D3: completeness 100% (Hassani²), 94.5% (Thomsen³), 98.6% (Vokuhl■). Domain D5: all studies had pre-registered protocols (IRCT, NCT05940649, NCT05997303). No study received "High risk" overall.

3.8 Certainty of Evidence (GRADE)

Table 9. GRADE Evidence Certainty Assessment

Outcome	Studies	Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	**Certainty**
Hypotension reversal (MAP target)	3 RCTs	RCT	–1 Some concerns	–1 Mixed outcomes/direction	None	–1 Small n (388)	⊕⊕■ Low
Haemodynamic stability (ARV-MAP)	1 RCT	RCT	–1 Some concerns	N/A	–1 High-risk only	–1 Single study	⊕⊕■ Low
Overshoot hypertension	2 RCTs	RCT	–1 Some concerns	None	None	–1 Wide IQR	⊕⊕■ Low
Arrhythmia	0	—	—	—	—	—	No evidence

Overall evidence certainty: **Low** (⊕⊕■).¹

4. Discussion

4.1 Summary of Principal Findings

This systematic review of three RCTs (n=388) evaluating push-dose NE for IOH in non-obstetric surgery provides several key findings:

1. Push-dose NE **10 mcg/bolus at 10 mcg/mL via peripheral IV** is the consistent dose and preparation across all included studies.²■
2. **Versus ephedrine**, push-dose NE is superior in MAP maintenance and reduction of hypotension episodes, with a more favourable heart rate profile.²

3. **Versus continuous NE infusion in low-to-moderate risk patients**, push-dose NE achieves comparable haemodynamic control with ~66% less drug consumption.³
4. **In high-risk patients** with continuous arterial monitoring, continuous infusion provides superior haemodynamic stability (lower ARV-MAP; large effect size).■
5. **Safety profile** is favourable: no significant overshoot hypertension, no documented arrhythmia, no peripheral vascular complications.²■■■

4.2 Push-Dose NE versus Ephedrine

The superiority of push-dose NE over ephedrine observed by Hassani et al.² is consistent with known pharmacological differences. Ephedrine acts through indirect sympathomimetic mechanisms, producing slower onset, greater heart rate elevation, and more variable response. Norepinephrine, acting directly on alpha-1 receptors, produces a more predictable vasopressor response with minimal chronotropic effect. The lower heart rate in the NE group (70.5 vs. 77.5 bpm, $p=0.012$)² is clinically significant, as tachycardia increases myocardial oxygen demand — a concern in surgical patients with potential coronary artery disease. These findings align with the broader shift in vasopressor preference from ephedrine toward alpha-1 agonists in the perioperative setting.¹

4.3 Push-Dose NE versus Continuous NE Infusion: The Role of Risk Stratification

The divergent results between the INDUCT trial³ (low-to-moderate risk, comparable AUC-hypo) and Vokuhl et al.■ (high-risk, inferior ARV-MAP with bolus) highlight that **patient risk stratification and BP monitoring modality** are critical determinants of optimal vasopressor strategy.

In the INDUCT trial,³ bolus NE was non-inferior to continuous infusion for AUC-hypo in routine elective surgical patients. For this broad population, a reactive push-dose strategy is adequate and has the added benefit of significantly lower drug consumption.³ In contrast, among high-risk patients with continuous arterial monitoring,■ the inherently reactive nature of bolus dosing generates greater beat-to-beat MAP fluctuation (higher ARV-MAP), which may be clinically important in patients with severe coronary artery disease, carotid stenosis, or recent stroke. The large effect size (Cohen's $d=1.00$)■ supports the clinical meaningfulness of this difference.

The scoping review by Berkenbush et al. (2024)■ contextualised the landscape of push-dose vasopressors and noted the paucity of high-quality evidence specific to the non-obstetric intraoperative setting — a gap that the present systematic review partially addresses through formal quality appraisal of the available RCTs.

4.4 Safety Considerations

Push-dose NE at 10 mcg/mL via peripheral IV demonstrated a favourable safety profile across all three studies.²■■■ Overshoot hypertension was not significantly different from comparators ($p=0.390$ ³; $p=0.986$ ■). No arrhythmia or vascular complications were documented, consistent with the low concentrations and doses used. The absence of structured arrhythmia outcome data remains a limitation — future studies should include formal cardiac monitoring.

4.5 Clinical Implications

Based on available evidence, the following clinical recommendations are supported:

1. **Push-dose NE 10 mcg** (10 mcg/mL, 1 mL bolus, peripheral IV) is a safe first-line vasopressor for acute IOH in low-to-moderate risk non-obstetric surgical patients. Repeat dosing every 2–5 minutes based on MAP response is appropriate.²■³
2. **Continuous NE infusion** is preferable in high-risk patients (ASA ≥III, major noncardiac surgery, significant cardiovascular comorbidities) with continuous arterial monitoring, for superior haemodynamic stability.■
3. **Push-dose NE is preferable over ephedrine** when avoidance of tachycardia and chronotropic effects is clinically important (e.g., ischaemic heart disease, severe hypertension).²
4. Peripheral IV administration of NE at 10 mcg/mL is safe for short-term intraoperative use based on current evidence.²■■

4.6 Limitations

1. Only three RCTs met inclusion criteria, limiting generalisability.
2. Significant clinical heterogeneity precluded meta-analytic pooling.
3. No study systematically measured arrhythmia or myocardial injury as structured outcomes.
4. ART-001 was conducted exclusively in hypertensive patients undergoing spinal surgery, limiting extrapolation to other surgical populations.
5. ART-002³ and ART-003■ focused only on the first 15 minutes post-induction; evidence across the full intraoperative period is lacking.
6. Assessment of publication bias was not feasible with only three studies.

4.7 Future Research Directions

1. **Multicentre RCTs** with standardised IOH definitions (MAP <65 mmHg for >1 min, per POQI consensus)¹ comparing push-dose NE vs. other vasopressors across diverse surgical populations.
2. Studies powered to detect differences in **arrhythmia, MINS, and AKI** as primary safety outcomes.
3. **Pharmacokinetic/pharmacodynamic studies** to optimise dosing regimens for different patient profiles.
4. Research in **resource-limited settings** where peripheral push-dose NE offers a practical, low-cost option.

5. Conclusions

Push-dose norepinephrine 10 mcg per bolus at a concentration of 10 mcg/mL via peripheral intravenous access is an effective and safe vasopressor for the management of intraoperative hypotension in adult patients undergoing non-obstetric surgery. It is superior to ephedrine in MAP maintenance with a more favourable heart rate profile, and comparable to continuous

infusion in low-to-moderate risk patients with substantially less drug consumption. In high-risk patients requiring continuous haemodynamic monitoring, continuous NE infusion provides superior haemodynamic stability. The overall certainty of evidence is Low (GRADE ⊕⊕■■), reflecting the small number of studies and clinical heterogeneity. High-quality multicentre RCTs with standardised outcome definitions are needed to strengthen the evidence base.

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